

Critical and Cultural Approaches to HCI

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Introduction

The field of human-computer interaction (HCI) has had to be as fast-moving as the technological changes that have marked the past few decades. Since the early 2000s, HCI entered its so-called 'third wave' (Bødker, 2006; Harrison et al., 2011; Kuutti, 2009), emphasizing non-workplace-based forms of computing, including ubiquitous, mobile, domestic, cross-cultural and entertainment computing and, more generally, experience design, among others.

Designing interactive software for these new contexts – the home, a village halfway around the world, the automobile – revolves around different problems from those of designing for professionals in the workplace. Consider the traditional goals of usability: low task completion time, low error rates, high learnability, high accessibility, and high subjective satisfaction (e.g. Gould & Lewis, 1985; Schneiderman & Plaisant, 2009). To what extent are these goals sufficient to guide the evaluation of massively multiplayer online role-playing games (MMORPGs), ebooks, and mobile apps? Presumably we want our games and apps to be usable, but we also want them to be fun, cool and exciting. It's fairly straightforward to measure error rates and task completion times, but how does one evaluate whether an interaction is 'cool' (Holtzblatt et al., 2010; Read et al., 2011)?

Some of the questions we now use to evaluate interactions are similar to the sorts of questions we use to evaluate art, literature, film, fashion or industrial products. HCI researchers also engage in ethical and political questions, for example, of designing for the developing world, confronting digital divides, and designing technologies for use in ecologically responsible ways. Not surprisingly, many researchers and practitioners in HCI have turned to design and cultural disciplines that have generations (and in some cases, millennia) of experience responding to such questions.

Within the broad field of HCI, a number of subdomains in particular have turned to and innovated on cultural approaches to HCI.

These include the following: experience design (Blythe et al., 2003; McCarthy & Wright, 2004), ubiquitous computing (Dourish & Bell, 2011), design-based approaches to HCI (Gaver et al., 1996; Löwgren & Stolterman, 2004; Zimmerman et al., 2007), reflexive HCI (Sengers et al., 2005), craft (Goodman & Rosner, 2011; Rosner & Ryokai, 2008), aesthetics (J. Bardzell, 2009, 2011; Fishwick, 2006; Udsen & Jørgensen, 2004), feminist and queer HCI (S. Bardzell, 2010; Light, 2011), HCI4D (Ho et al., 2009), affective computing (Boehner et al., 2007), intimate and

sexual interaction (Bardzell, J. & Bardzell, S. 2011; Kannabiran et al., 2011; Kaye, 2006), and domestic computing (Bell et al., 2005), among others.

The goal of this chapter is to offer an overview account of what cultural theory is; to explore cultural theories' intellectual practices, domains and concerns; and above all to explore both existing and as yet unexplored linkages between cultural scholarship and HCI, with an eye to enriching the development of HCI's third wave.

Knowledge and/of Culture

The idea of 'culture' is famously difficult to define. Rather than offering my own attempt at a definition, I will let those in cultural studies speak for themselves. Two introductory books on cultural theory (Milner & Browitt, 2002; Smith, 2001) begin with definitions of culture based on the seminal work of cultural theorist Raymond Williams. Williams, as paraphrased by Smith, offers three common understandings of culture.

- To refer to the intellectual, spiritual, and aesthetic development of an individual, group, or society.
- To capture a range of intellectual and artistic activities and their products (film, art, theatre). In this usage culture is more or less synonymous with 'the Arts,' hence we speak of a 'Minister of Culture.'
- To designate the entire way of life, activities, beliefs, and customs of a people, group, or society. (Smith, 2001: 2)

These definitions collectively suggest that culture refers to the world of ideas and their expressions in a given social group. In Milner and Browitt's formulation, culture refers to 'that entire range of institutions, artefacts and practices that make up our symbolic universe' (Milner & Browitt, 2002: 5). Culture can be understood in terms of an intersubjectively participatory inner life that is simultaneously constituted by and given expression through symbolic systems (or 'texts' as they are often called, though they may be novels, films, paintings, or even advertisements, comic books, and other forms of popular culture).

This formulation of culture has implications for how researchers study it, a point that is clearest when set in contrast to other scholarly approaches. For example, much knowledge in the natural sciences is derived from the discovery of facts in the physical world, for example the chemical composition of different types of volcanic material, the mating behaviors of a certain kind of animal or patterns of celestial movements. Much work in the social sciences seeks similarly to discover facts of the human social world, such as the relationships between organizational structures and the creative output of professional teams, the effects of different

pedagogical strategies on fourth grade reading scores or relationships between economic incentives and social behavior.

In contrast to science's emphasis on the *discovery* of facts, humanistic questions are often not so much fact-based as they are based in rational accounts of the *judgments* we form as to whether a work is good, what it means, how its formal qualities bear certain meanings, how innovations in production techniques evolve an expressive symbolic vocabulary, how a work reflects and comments upon the circumstances under which it was made, and so forth.

Several philosophers and cultural scholars (e.g. Barnard, 2001; Taylor, 1971) argue that knowledge of cultural phenomena is of a different kind from the knowledge of the natural world and, accordingly, the relationship of the knowing subject (i.e. the scientist, the cultural critic) to the objects of enquiry also differs. Another way to express this idea is that scientific and humanistic disciplines tend to prioritize different epistemologies. An *epistemology* is a theory of knowledge, that is, an account of how knowledge is acquired, how it is evaluated, what doesn't count as legitimate knowledge (e.g. the rejection of astrology as pseudo-science), who is able to acquire or create knowledge and under what circumstances, etc.

One challenge for HCI has been that the predominantly science-trained HCI community has not always been in a strong position to perceive or evaluate the rigor or quality of humanist contributions to the field. In other words, humanistic research runs the risk of coming across as bad science because its knowledge practices and forms of rigor differ. I hope this chapter can contribute to bridge building within HCI between scientific and humanistic knowledge traditions. In the following sections, I will first elaborate on ways that humanist and scientific epistemologies differ, then I will elaborate on ways that they are similar.

Scientific Versus Humanistic Epistemologies

Malcolm Turvey, a philosopher of film, explores the differences between scientific and humanistic epistemologies in an essay in which he challenges the fitness of scientific models for the study of film. Turvey offers a summary of how empirical research and the development of theory have been so successful in science, then he points out that film theorists do not generally collect any data or attempt to verify empirically their own claims. In light of all this, Turvey wonders, 'how is it that theories about [films] ever convince anyone...?' (Turvey, 2005: 25).

His answer to this question is, whereas science seeks to discover laws and patterns that are hidden from everyday observation, 'film theories concern *what human beings already know and*

do' (2005: 25, italics in original). For example, we already know what a romantic comedy is, and we know how to respond to it, and how to behave in the theatre during it; all of this is present to everyday observation and awareness and need not be discovered. The problem is that our awareness of such norms is fuzzy: we often sense, for example, that a film, novel, or song is beautiful or resonates with us in some very personal way, but we struggle to explain why or how. We have all been in a museum or a class and experienced that moment of revelation when the docent, or, teacher points to a detail, a phrase or a stroke and tells us what it means (e.g. symbolically, its historical significance, how it was made, etc.). That detail had been right in front of our eyes all along, and we needed no microscope or DNA test to see it, yet we just overlooked the detail and its significance; now, we can't see the work without focusing on that detail. The ability to perceive such details and relate them to an account of a work's overall meaning is one of the paradigmatic skills of the humanities scholar. As aesthetic philosopher Marcia Eaton puts it, critics 'are in the business of *pointing*. They get us to *perceive* some features of an object or event' (Eaton, 1988: 116).

The ability to perceive and point to the right details is not something one is born with, since it depends on one's awareness of norms and conventions, that is, things we learn throughout life. Alexander Baumgarten, the eighteenth-century philosopher who first coined the term 'aesthetics', defines it as 'the perfection of sensory cognition', a human faculty that includes keenness of sensation, imaginative capacity, penetrating insight, good memory, poetic disposition, good taste, foresight and expressive talent (Shusterman, 2000: 264–65). This faculty is cultivated or improved through years and years of practise by engaging both canonical texts (again, 'texts' is used in an inclusive sense) and generations of scholarship on and criticism of those texts.

The humanist epistemological foregrounding of perception, insight, imagination, etc., has strong implications for what constitutes good humanist knowledge. A successful critical interpretation, in the words of philosopher Richard Eldridge, captures 'genuine aspects of a work's meaning, of how [the work] distinctly presents a subject matter as a focus for thought and emotion, fused to the [critic's] imaginative exploration of material' (Eldridge, 2003: 147). Eldridge's statement here nicely summarizes humanistic attitudes towards the object of their enquiry (i.e. a text that 'distinctly presents a subject matter as a focus for thought and emotion') and the subject of enquiry (i.e. the scholar-critic engaged in an 'imaginative exploration of material'). 'Text' and 'data set' are two different kinds of objects of enquiry, and the critic and the objective scientist are two different kinds of knowing subject.

The Commonalties of Scientific and Humanistic Work

In the preceding section, I drew contrasts between scientific and humanistic ways of knowing. Drawing such contrasts is constructive because it helps readers understand the different intellectual problems, goals and priorities of scientific versus humanistic disciplines. The disadvantage of drawing such contrasts is that it seems to suggest there is an unbridgeable gulf between science and the arts, as if they are completely incommensurate. However, I do not want to reinforce a ‘two cultures’ narrative, especially when I am advocating that scientific and humanistic approaches be used together in HCI! (It is also worth noting that some academic fields – anthropology in particular – have productively blended these epistemologies for decades.)

The prioritization of perceptiveness and keenness of insight in humanist works should not imply that scientific works lack any of these. The distinctions drawn here between science and humanistic approaches have to do more with their *primary* epistemologies, as opposed to their whole processes or products.

In other words, the primary means by which knowledge is achieved may differ for scientific versus humanistic disciplines, but all disciplines nonetheless display qualities of the other. For example, a rigorous empirical study has critical or humanistic aspects inasmuch as its researchers make decisions or judgments about what data and in which domain they should conduct their research as well as in making decisions about how to interpret their data and present it to their peers. Submitting a successful scientific paper also requires a mastery of the norms of a given journal's or conference's readership. Likewise, a rigorous humanistic project involves discoveries of fact: what is in a given text, the conditions of its production, and the history of its reception. Understanding a cultural text also entails the development of generalizations based on evidentiary particulars, and such understandings are evaluated by the community in part on their ability to explain those particulars.

Philosopher of science Dagfinn Føllesdal points out a more fundamental similarity between humanistic and scientific scholarship. Both, he claims, use the *hypothetico-deductive method*. This method entails ‘an application of two operations: the formation of hypotheses and the deduction of consequences from them in order to arrive at beliefs which – although they are hypothetical – are well supported’ (Føllesdal, 1994: 234). Individual hypotheses have credibility inasmuch as they are both logically consistent and fit with our experience. Also, a collection of related hypotheses is compelling inasmuch as it achieves theoretical simplicity. From such a perspective, we might say that the following two claims are equivalent.

- Hypothetical inferences derived from biological data can be explained by the theory of evolution.

- Hypothetical inferences derived from Federico Fellini's films can be explained with the theory that Fellini's Italian Catholic upbringing deeply affected his views of the world, his sexual relationships, and himself.

From the perspective of how we reason about our objects of enquiry, scientific and humanistic approaches aren't so different after all. Neither is merely opinion-based, and both produce claims that have an evidentiary basis that is persuasive to an intellectual community.

An Example: The Science and Culture of User Experience

The broad-brush philosophical summary I have just offered between scientific and humanistic epistemologies might come across as irrelevant to HCI. Yet recent developments in the field are recapitulating the types of issues described here.

One of the most visible examples is (what is in my view) HCI's schizophrenic take on user experience, which can be summarized as follows.

- One body of UX research treats user experience as a discoverable phenomenon, out there in the world. This broadly scientific approach uses theories, such as Csikszentmihalyi's 'flow', and diverse methods to discover it, for example self-report techniques using the Positive and Negative Affect Schedule (PANAS), questionnaires with Likert scales, and the use of physiological sensors to detect user engagement, frustration, and other experiential qualities. Examples of this kind of user experience research include Hassenzahl et al. (2010), Law et al. (2008), Law et al. (2009), Liao et al. (2006), Picard (1997), Picard and Klein (2002) and Sutcliffe (2010).
- Another large body of research treats experience as a form of felt life to be interpreted. This broadly cultural approach leverages philosophy, literary theory and design methods to help the UX researcher critically interrogate a user experience situation. Examples of this kind of user experience research include McCarthy and Wright (2004), Buchenau and Fulton Suri, (2000), Forlizzi and Battarbee (2004), Nardi (2010) and Boehner et al. (2005).

The HCI community accepts user experience (UX) research contributions from both scientific and cultural approaches, which I regard as a very positive development. Less encouraging is the fact that these two literatures on UX seem to be developing almost in isolation of one another: references from one side to the other are few, and when they do occur, they are often unsympathetic (Bardzell, 2012).

If we want to build links between these two approaches, we might begin by observing that UX research seems to involve two different underlying questions.

1. What do actual users like about, and how do they feel about, given experiences involving interaction design?
2. How can user experience researchers and designers perceive what is most worthy of perceiving in an experience involving an interaction design?

The first question is fundamentally empirical in nature: the desired information is ‘out there’ (i.e. users do have preferences, feelings and perspectives about their experiences), and the job of the UX researcher or designer is to discover that information in the best way possible. The second question is about distinguishing relevant elements of an experience to perceive and to clarify their significances and meanings. For example, a film critic can identify links among editing or *mise en scène*, broader cinematic or cultural movements, available production technologies and techniques, and experiential qualities. By the same reasoning, an interaction designer should be able to ‘read’ relationships among the visual languages of an interaction, its functionality, and the most important use qualities that contribute to its experience.

Is there any doubt that competence in answering both questions would support the design of better interactive experiences? Also, answers to each of the two questions should be able to inform the other. A rigorous empirical data set could effectively expose issues of interest – aspects of an experience which were, for example, surprisingly good or bad – that a more cultural perspective could then be leveraged to interpret in a more culturally sensitive and perceptively insightful way.

Conversely, more critical understandings of how experiential qualities arise from interacting with certain types of systems and interfaces, given relevant socio-technical norms surrounding such interactions, could lead to innovation in the development of empirical measures and research instruments. It seems that the HCI community is ready for both cultural and scientific approaches to coexist, not just in experience design but in any number of areas (e.g. affective computing, embodied interaction, HCI4D, etc.). Yet if each of these topics replicates a ‘two cultures’ pattern, we will have missed out on the potential benefits of a more synergistic approach. My hope is that a better understanding of humanist scholarly norms in the HCI community will help prevent excellent critical work from looking like it is just bad science and hopefully decrease some of the confrontational rhetoric that sometimes occurs, while, ideally, stimulating mutually beneficial collaborations among researchers and practitioners from both intellectual traditions.

Doing Cultural Hci

For the balance of this chapter I will focus on ‘doing cultural HCI’ as a knowledge practice. One

aspect of this is the core competencies or foundational abilities of cultural approaches to HCI, including design thinking and criticism. Another aspect is how critical and cultural theory has moved into HCI practice. I explore each of these in turn.

Developing an Ability to Do Cultural HCI

In both the design and humanities literatures, the designer or critic is understood to be a special kind of person, what I characterize as an expert subject. By *expert subject* I refer to people who, like master crafters or important filmmakers, are better than the average person at what they do, and yet importantly they are better at it in a very individual or personal way; for example, a Raymond Loewy design or an Alexander McQueen dress. This means that the products of their efforts – be they designs, crafts or essays – are recognized as valuable to a community but also recognized as unique to them, that is, not replicable.

The value that design brings is not a fact to be discovered. As Nelson and Stolterman describe it:

The success of the design process can best be determined when those being served experience the *surprise of self-recognition*. This comes when that which emerges from a design process meets and exceeds the client's original expression of that which they (usually only dimly) perceived as desirable in the beginning. This original expression of what is desired is known as the client's *desiderata*. The designer's role is to midwife that desiderata, which could not have been imagined fully from the beginning, by either client or designer... (2003: 48)

The key vocabulary in this quote – perception, imagination, and the metaphor of midwifery to describe the collaborative labor of bringing something new of value into the world – unmistakably echoes the notions of aesthetics as ‘the perfection of sensory cognition’ described earlier in this chapter. The role of the designer is to serve as the expert subject who creatively and insightfully perceives and pursues solution opportunities within the desiderata. This subjective expertise is sometimes referred to as *design thinking*, that is, the idea that professional designers share a knowledge practice, a certain way of reasoning about design problem domains, solution generation, and solution evaluation that enables them to be skilled designers (Cross, 2007; Rowe, 1987).

This thinking is not a rare cognitive faculty that only some people get to have: design thinking is an outcome of training. In design school, students are immersed in a design culture and gradually learn to tacitly appropriate its knowledge and skills (Snodgrass & Coyne, cited in Greenberg & Buxton, 2008). Löwgren and Stolterman describe the cultivation of design ability

through, among other activities, developing a 'sense of quality', a position that clearly resonates with the notion of criticism as pointing and as a form of skilled perception, as described earlier. This sense of quality includes an insightful appreciation of and ability to create quality, and it 'has to be developed, continuously challenged, and improved' (Löwgren & Stolterman, 2004: 58).

A primary way that designers and scholars of culture develop, continuously challenge and improve this ability is through criticism. *Criticism* refers to expert acts of judgment, that is, the practice that leads to the acts of perceptive pointing and the clarifying of norms and meanings that I described earlier in this chapter. It is often motivated by vague yet intriguing intuitions that we have when we respond to human works. Philosopher of film Stanley Cavell (Cavell & Klevan, 2005: 181) describes how certain moments in a film speak to him in this way: 'I didn't only think the shots were unusual, or striking, I thought they were gently mysterious, and that they were significant. They asked questions of me'. Once certain moments in a film capture his attention, he then rationally interrogates them to try to understand why they affected him in that way; he tries to solve their gentle mystery. Schön and other design theorists also have described this idea of designs and materials speaking back to them (Schön calls it 'backtalk'). In this way, the framing of problems and the proposing of design solutions happen iteratively in a 'dialogue' between designers and their materials and designs.

Summarizing, criticism can be characterized as the rational investigation of wisdom or intuition, initially enabled by a receptivity to or perceptiveness with regard to how human works speak to us and then developed through learned interpretative strategies made possible through exposure to comparable examples, theory and knowledge about the contexts of creation and experience. This rational investigation clarifies the relationships between objects and their subjective experiences; intensifies and beautifies those experiences (Dewey, 1934; Shusterman, 2000); and even reveals the false pleasures of those experiences, that is, the ways in which works are 'subjecting you to false views of yourself and of the world' (Cavell & Klevan, 2005: 178). Critical interpretation and backtalk in this sense is also inexpensive (compared, at least, to scientific enquiry) and practised daily; my experience of working with designers is that they critically interpret designs more or less constantly and reflexively, both at work and in everyday life.

In my own work, I have sought to explore criticism as it applies to interaction design (Bardzell, 2009, 2011). In Bardzell (2011), I define *interaction criticism* as

rigorous interpretive interrogations of the complex relationships between (a) the interface, including its material and perceptual qualities as well as its broader

situatedness in visual languages and culture; and (b) the user experience, including the meanings, behaviors, perceptions, affects, insights, and social sensibilities that arise in the context of interaction and its outcomes.

I have argued that in critical fields – literature, art, design, philosophy, media studies – these interrogations tend to emphasize one of the following four fundamental perspectives.

- *The creator and the act and situation of creation.* Critical reasoning from this perspective foregrounds creative agency, including creative intentions, skills, desires and influences. When we talk about ‘a typical Hitchcock film’, we are reasoning from this perspective. Studies of designers, design processes, crafters, and the notion of a designer's unique signature or voice also relate to this perspective.
- *The artifact itself.* This perspective focuses on qualities of the artifact, including its content and arrangement in form, its material qualities, and its visual language. When we think about how a work references other styles, genres or movements, we are reasoning from this perspective. Much traditional design criticism emphasizes this perspective, and HCI would do well to develop this kind of critical vocabulary more explicitly for interaction.
- *The individual consumer, reader, or user and the experience.* A work is only meaningful when it enters into human experience, including perception, imagination, emotional reaction, and so forth. When we talk about the emotional qualities or meanings of works, we are generally speaking from this perspective. User experience research is often situated in this perspective.
- *The socio-cultural context.* Works are created, distributed and become meaningful not just in individual minds but also more broadly in socio-cultural contexts. National, political, economic, gendered contexts influence meanings and perceived value, often in ideological ways. Gender and HCI, HCI4D, and sustainable interaction design are a few areas that are increasingly exploring HCI from this perspective.

I want to stress from the outset that we all use these four perspectives all the time. Everyone at one point or another has chosen to watch a movie based on the director, or on its a genre (e.g. horror) in anticipation of having a certain kind of experience, or has chosen a movie because it reflects or challenges their politics or values. Everyone has tried to interpret what an author or painter might have meant in a novel or painting. We can all easily recognize gender stereotypes in old advertisements. Again, cultural approaches are about clarifying what we already know and do; I view them as different in degree, not in kind, from the everyday critical judgments we all make.

What cultural scholarship adds to this are thousands of theories and concepts applied to works from nearly every culture. Such theories offer models that support our attempts to rationally

interrogate our intuitive reactions from each of these perspectives more rigorously and insightfully. The challenge for interaction design professionals thus lies in choosing and leveraging such theories to cultivate their intuitive reactions into intellectually useful design understandings.

From Cultural Theory to HCI Practice

In the previous section, I sketched some of the background ability common to arts and humanities-influenced knowledge workers, including designers. I stressed that the lifelong practice of criticism helps with the cultivation of a perceptive and imaginative eye, and I offered some broad perspectives commonly used to frame such activities. Yet such descriptions fall far short of a methodology. How does cultural theory translate into HCI practice?

If we understand a *methodology* as the link between an epistemology (i.e. a theory of knowledge-making) and individual research methods, then a methodology would seem to be just what we need to connect the rich theoretical tradition of the humanities with the practice-oriented demands of interaction design. However, the notion of a ‘methodology’ is seldom used in the humanities, which emphasizes instead the creativity and insightful perception of the individual critic over any replicable approach. I do not mean to imply that linking cultural theory design practice is impossible; it just suggests it may be a little bit difficult and it will take some work. In the following three subsections, I summarize instances where this has occurred, beginning with one or more critical theories from the humanities, moving into HCI by way of theory, and from there influencing HCI research and practice methodologically.

Aesthetic Experience: From Dewey to HCI

The first example involves the work of American pragmatist philosopher John Dewey, who wrote extensively on art and experience in the middle of the twentieth century (Dewey, 1925, 1934). Concerned with the ways that industrialization and factory labor created deadening environments, Dewey sought to improve the quality of life for people through better education and a more just society. These philosophical explorations led to one of his most enduring contributions: a theory of aesthetic experience. In contrast to traditional notions of aesthetic experience, which centered on the appreciation of fine arts by the leisured classes or limited them to museum experiences for everyone else, Dewey sought to articulate a notion of everyday aesthetic experience.

A very basic distinction in Dewey's analysis is the difference between the constant flow of experience and ‘an experience’, that is, a grouping of experience-flow together in the form of a coherent entity, for example a memory of an experience or the story one tells of an experience.

Dewey wants to understand what makes a good 'an experience', and he proposes a model framework. As a coherent entity, an aesthetic experience has a beginning, a middle and an end. Each 'an experience' is directed towards some end or purpose; that is, it has directedness. Within this directed experience, there is 'doing' and 'undergoing', that is, actions by and reactions to the experiencing subject. These doings and undergoings are characterized by variety and movement, which are pleasurable. All of this leads to a consummation, which satisfactorily concludes the aesthetic experience.

Dewey's work has been picked up in HCI and design fields several times. Donald Schön's dissertation was on Dewey, and design theorist Erik Stolterman has suggested that Dewey's thought pervades Schön's *The Reflective Practitioner* (private communication). In HCI, McCarthy and Wright (2004) wrote a highly influential book on experience design, *Technology as Experience* (alluding to Dewey's book, *Art as Experience*). McCarthy and Wright apply Dewey's theory of experience within the experience design domain. In doing so, McCarthy and Wright propose norms by which researchers, practitioners and users alike might evaluate the experiential qualities of interaction. Recalling how humanistic scholarship tends to stress the clarification of norms surrounding what we already know, it is easy to see McCarthy and Wright's work as fitting squarely in this tradition. Their application of Dewey offers a theory that makes UX practitioners and researchers more perceptive and imaginative in their engagement with users, interaction designs and experiences. Thus, one HCI outcome of an engagement with cultural theory is the emergence of new HCI theory.

A more recent example in HCI is Bonnie Nardi's (2010) use of Dewey's theory of aesthetic experience in her ethnographic work on World of Warcraft (WoW). Nardi applies three theoretical frames – Skinnerian behavioralism, Deweyan aesthetic experience, and activity theory – to construct an understanding of the ways in which WoW play is pleasurable. Nardi takes on issues that resonate with the WoW player community, such as the buying and selling of player accounts, and constructs an insightful understanding of them using these theoretical constructs. In effect, her work is strikingly similar to Cavell's process of moving from initial intuition to theoretically informed rational interrogation. Nardi's use of Dewey suggests another HCI outcome of engaging with cultural theory: a critical interpretation, or 'reading', of a complexly layered, yet important interaction in its particular context.

From Critical Theory to Critical Design

A common goal for most strands of critical theory is an effort to use theory to reveal hidden, yet important, aspects of culture, and through that revealing, to subject them to intentional change. In this regard, critical theory and design seem like natural partners. This linkage has been

explored in critical design (Dunne, 2006; Dunne & Raby, 2001). Dunne offers the following definition of critical design: It is:

a design approach for producing conceptual electronic products that encourage complex and meaningful reflection on inhabitation of a ubiquitous, dematerializing, and intelligent environment: a form of social research to integrate critical aesthetic experience with everyday life.... I hope in my approach I have retained the popular appeal of industrial design while using it to seduce the viewer into the world of ideas rather than objects. Industrial design locates its object in a mental space concerned with identity, desire, and fantasy and shaped by media.... Again, I hope this remains intact but is subverted to challenge the aesthetic values of both consumers and designers. (2006: 147)

Critical design promises an approach that, rather than serving needs as they are presently understood, instead seeks to disrupt, to transgress such constructions of need and encourage people to 'enrich and expand our experience of everyday life', to stage dilemmas 'that force a decision onto the user, revealing how limited choices are usually hard-wired into products for us' (Dunne & Raby, 2001: 45–6). Examples of works that feature critical designs are Gaver et al. (2004, 2010) and Boehner et al. (2005).

Critical design is 'critical' in the sense that it foregrounds perception and imagination, rather than the discovery of facts and their translation into needs and requirements. By challenging users' expectations, critical designs invite users and researchers to perceive and reason about norms in new ways. As Cavell – and generations of Marxist and feminist cultural researchers – have warned, art and design can also provide false pleasures, by entertaining us or serving our functional needs in a way that gives us a false or unjust understanding of the world: critical designs can subversively challenge these false pleasures. By provoking those involved into discussion, critical designs also stimulate an imaginative discourse that is oriented towards the articulation of new desiderata, simultaneously leading to new designs, but also, more radically, leading to demand for new kinds of design. At the same time, critical design strives to remain grounded in everyday life; Dunne (2006: 84) writes, 'But if [critical design] is to avoid accusations of escapism this design thinking must also develop strategies for linking itself to everyday life'.

Cultural HCI Methodological Innovations

Interaction criticism and critical design are not the only methodological innovations that have origins in cultural HCI. I will sketch four more in this section and then explore their underlying commonalities.

The first is ‘pastiche scenarios’, as presented in Blythe and Wright (2006). The basic idea behind pastiche scenarios is to blend together the conventional HCI notion of a scenario with a character or scene from a literary narrative, such as a novel. Whereas the goal of a traditional scenario is to put a human face and everyday situation on empirical data sets, the goal of a pastiche scenario is to interrogate the felt life of interaction in much richer ways – albeit at the expense of strict fidelity to the data itself. For example, the authors used Bridget Jones to construct a pastiche scenario about iPod use, not only because she meets the core demographic criteria of iPod users, but also because ‘she brings more to the scenario than her age, occupation, and gender’, including a ‘distinctive narrative voice’ and very specific habits, such as her anxieties about how she is perceived, her habits of linking specific songs to memories of prior boyfriends, and a general sense of disorganization in her everyday habits (Blythe & Wright, 2006: 1146). Pastiche scenarios push the underlying logic of personas and scenarios – to put a human face on data to help us understand it – to another level, one of creative empathy with the felt life of others.

Another methodological innovation is the concept of *defamiliarization*, introduced in Bell, Blythe and Sengers (2005). The concept is originally derived from literary theory, specifically the work of the Russian formalists in the early twentieth century, who argued that the *telos* (or end) of art ‘is to change the mode of human perception, to render [hitherto] imperceptible formulas [so that they become] unusual and palpable’ (Steiner, 2005: 18). In their paper, Bell, Blythe and Sengers (2005) describe one of the chief difficulties in designing new technologies for the home as the fact that the home is so overly familiar to us it is hard for us to see it critically. They offer defamiliarization as a methodological approach to dealing with this problem. One strategy is to use an historical and cultural analysis of the home, to identify historical trends – such as the rise of industrialized efficiency in the kitchen, which is not combined with a rise in the quality of food – and situate them alongside emerging designs in HCI as a means to exposing their assumptions. Another strategy is to study homes in different countries; studies of homes in Asia helped the researchers perceive numerous differences in the ways families use them and how family members relate to each other within their homes.

Comment [RC6]:

Not in Refs?

The third methodological innovation I wish to introduce is *feminist HCI*, which seeks to bring together feminist theory and HCI research and practice. In a recent paper, Shaowen Bardzell

(2010) introduced the term and then motivated and defined it with an interesting rhetorical strategy. She began by introducing feminism itself – its core values, positions and epistemologies. Next, she explored its influence in design fields that are comparable to HCI, including industrial design, urban planning, game design. Next, she synthesized research in HCI that revolved around key themes of feminism, even though it did not use that vocabulary; she then turned her synthesis into an analysis of opportunities for feminism to contribute to HCI, before concluding with a handful of qualities of feminist HCI. Her technique of tracing feminism from its origins in critical theory through other design fields and into HCI is perhaps the most explicitly genealogical importation of cultural theory into HCI. Building on this work, we subsequently proposed an outline of a feminist HCI methodology that offers concepts to support researchers' navigations between the scientific imperative to 'know truth' and the moral imperative to 'do good' (Bardzell, S. & Bardzell, J. 2011).

The final methodological innovation I will describe here is slightly different in character but equally important. It is a particular form of analysis developed by Phoebe Sengers and her colleagues, in particular in Boehner et al. (2007) and DiSalvo, Sengers and Brynjarsdóttir (2010), which Sengers calls the *epistemological survey*. An epistemological survey collects a corpus of the works in HCI on a given topic; it then *critiques* that corpus to expose the unsaid, the assumptions, and even the ideologies implicit in the corpus. In DiSalvo, Sengers and Brynjarsdóttir (2010, 1976), for example, they asked the same set of questions of each of the roughly 70 papers in the corpus.

(1) how does the paper define and justify attention to sustainable HCI? (2) what disciplinary orientation is used? (3) how is the problem of sustainability and its solution framed?... (4) How is the role of the researcher framed? (5) Who takes action, or is supposed to take action? (6) Who is considered the 'expert,' and whose point of view is questionable? (7) How do the authors deal with political disagreements about the environment? (8) Does the paper aim to establish a definitive truth, or does it leave open the possibility of serious differences of opinion about its subject matter? (9) What constitutes success?

One of the primary goals of this critical analysis is to explicate the subject positions that are constructed through the corpus. Who is the researcher? Who gets to act? Who is framed as part of the problem? If design is about effecting change, and change is inherently political, then it would follow that HCI is a political field. However, HCI does not always acknowledge its own political dimensions, the ways that it asserts power, introduces divisions, or establishes who is helping and who is fixing whom. As the authors write, 'Our goal is to provide a reflective lens for

practitioners of sustainable HCI which will allow for principled, reflective discussion of how we have, until now, defined sustainable HCI, and how we might best choose to do so in going forward' (DiSalvo et al., 2010, 1975) It is worth taking a moment to summarize some features that are common across two or more of these works. The most basic feature is the use of critical theory and its approaches in HCI; that much is obvious. Another feature that they share is a notion of the researcher as an expert subject, someone who is responsible for creative perception and insightful interpretation of some aspect of HCI. The pastiche scenarios paper relies heavily on the researcher's ability to identify an appropriate literary source and to apply it thoughtfully. The notion of reflexivity (Sengers et al., 2005) is also crucial to this work: critical HCI researchers do not only see themselves as incrementally adding knowledge to the literature that has come before but also, significantly, as re-evaluating that literature and redefining ourselves as a result. Bardzell's feminist HCI paper suggests that much HCI research was already feminist, even though that term ('the F-word' as one attendee at her talk put it) was not used: in this sense, feminist HCI brought about a 'surprise of self-recognition'. The methodologies described here are also historically and/or spatially situated; for example, the defamiliarization paper uses history and different countries as broad strategies to defamiliarize the here and now. Another feature common to these methodologies is a political awareness, which is made explicit in both the feminist HCI and the epistemological survey research.

Conclusion

I have introduced culture, cultural scholarship and the application of cultural scholarship in design and HCI. Though it is clear that HCI wants to (and must!) understand culture better, I want to conclude this chapter by stressing that cultural HCI should have less to do with *telling us about* culture and more to do with *helping us improve* culture. It would be wrong, I argue, to see cultural approaches primarily as another research lens to tell us what is out there in the world; the social sciences are a better fit for this direction of enquiry. Cultural approaches should be used to help HCI improve our lived environment and improve ourselves. I see two broad strategies for pursuing these goals.

First, HCI should orientate itself towards the production of technologies and interactive experiences that are of genuine and lasting value. That means our work should meet 'human needs' in the broadest, deepest and most robust sense possible: social, emotional, physical and spiritual needs at both individual and societal scales. We should contribute to the making of artifacts, spaces, technologies, practices and social arrangements that challenge and ennoble us to be more empathic and more vigilantly participatory and democratic. We should be careful that we are not contributing to false pleasures. This all sounds like a tall order, but I

do not propose it as a literal agenda – rather, as an orientation to our research agendas; let this, not efficiency and productivity, be our ultimate goal.

Second, *HCI – like the best art, craft, and traditional design – should cultivate us as appreciators of what is worthy*. This includes the education of our senses and imagination, as proposed in the eighteenth century by Baumgarten and developed more recently by Shusterman. It includes increasing our capacity for empathy, which philosopher Richard Rorty suggests can be achieved through our encounters with literature and art – why not interaction design? It includes contributing to community aesthetics at the local, national, and even global levels, because what people appreciate, they want. As we saw with critical design, helping users perceive the world more freshly, provoking and surprising them, can spur both imagination and demand, improving quality of life today and generating the desiderata for a better tomorrow. The philosopher Michel Foucault suggests that each of us ought to treat our self as a work of art. Let us HCI researchers and practitioners challenge ourselves to be a part of that project.

References

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